Question 11 – 25 Marks

Technological Advancements Behind Bitcoin (2009)

- Bitcoin was created in 2009 by Satoshi Nakamoto.
- It combines multiple existing technologies into one system.
- These technologies together enabled a decentralized digital currency.

1. Cryptography

- Public and private key cryptography secures ownership.
- Users sign transactions using private keys.
- Only the owner can spend their Bitcoin.
- Based on work by Whitfield Diffie and Martin Hellman.

2. Hash Functions

- Bitcoin uses SHA-256 hashing algorithm.
- Hashes link blocks together securely.
- Any small change changes the entire hash.
- Based on work by the NSA.

3. Peer-to-Peer (P2P) Networking

- Bitcoin runs on a peer-to-peer network.
- No central server or authority exists.
- All nodes share and verify transactions.
- Inspired by BitTorrent and earlier P2P systems.

4. Merkle Trees

- Merkle Trees efficiently store transactions in blocks.
- Only one hash is needed to verify many transactions.
- Proposed by Ralph Merkle.

5. Distributed Ledger

- The blockchain is a shared public ledger.
- Every node has a copy of the ledger.
- Ensures transparency and trust.

Conclusion

- Bitcoin is not a single invention.

- It is a smart combination of existing technologies.
- Satoshi Nakamoto successfully integrated them into a working system.

Question 12 – 25 Marks

Proof of Work (PoW) Consensus Mechanism

- Proof of Work is used by Bitcoin to achieve consensus.
- It ensures agreement on valid transactions.
- Miners compete to add new blocks.

How Proof of Work Works

- Transactions are grouped into a block.
- Miners try different nonces to find a valid hash.
- The hash must meet network difficulty rules.
- First miner to solve it adds the block.
- Miner receives block reward and transaction fees.

PoW Difficulty Adjustment

- Bitcoin adjusts difficulty every 2016 blocks.
- Keeps block time close to 10 minutes.
- Prevents too fast or too slow mining.

Security Benefits of PoW

- Very hard to change past transactions.
- Requires massive computing power to attack.
- Protects against double spending.

PoW vs Proof of Stake (PoS)

- PoW uses miners and electricity.
- PoS uses validators and staked coins.
- PoS is more energy efficient.
- PoW is considered more battle-tested.

Recent Events Related to PoW

- Concerns about energy consumption.
- China mining ban in 2021.
- Growing use of renewable energy.
- Ethereum moved from PoW to PoS.

Text Diagram – Proof of Work

Transactions

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Block Creation

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Miners Solve Hash Puzzle

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Valid Hash Found

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Block Added to Blockchain

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Reward Given to Miner